LipScribe — Milestone & Progress Report Data

Source: Trello board "LipScribe — CVDL Group 4" (https://trello.com/b/3QFTxC29/lipscribe-cvdl-group-4)
Snapshot taken: Thursday, 29 October 2026 (~10:15 UTC)
Reporting window: 19 Oct – 29 Oct 2026 (board created Oct 19; covers Week 1 and Week 2 of a 6-week plan)

> Caveat: most cards were moved to Done in bulk on Oct 29, so Trello dates show when the board was updated, not necessarily when the work finished. Treat completion dates as "recorded on" dates.

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1. Project summary

LipScribe is an audio-visual speech recognition system built on the GRID corpus. It combines a visual (mouth-crop) stream and an audio stream through cross-modal attention and gated fusion with modality dropout, and decodes with CTC. It is evaluated at five SNR conditions: clean, 10, 5, 0 and −5 dB.

Experiment IDs

ID	Description
B1	Audio-only baseline
B2	Visual-only baseline
B3	Concatenation-fusion baseline
P	Proposed: attention fusion + modality dropout
X1	P without modality dropout
X2	P with concatenation in place of attention (keeping dropout)
X3	P with 2D per-frame CNN in place of 3D front-end
X4	P with EfficientNet-B0 trunk (optional)
X5	B1 with MFCC / log-Mel+MFCC features
X6	P with SNR auxiliary head (optional)

Drop order if schedule slips (REF-03): optional extensions → X4–X6 → live Mode B (Mode A stays) → multiple seeds. Never dropped: B1–B3, P, per-SNR evaluation, X1–X3.

2. Team and roles

Member	Role
Charis Opol	Project coordinator and training lead (board, reviews, train.py, training runs, GPU booking, submission)
Cole Rodney Migisha	Model lead (mouth cropping, visual/audio encoders, attention fusion, modality dropout)
Ainomugisha Faith	Data lead (GRID, integrity checks, splits, manifest, crop extraction, augmentation, ablations)
Lisa Lusinga Namugenyi	Audio and evaluation lead (MUSAN, noise mixing, tokeniser, evaluate.py, stats, report editor)
Wakisa Isabelle	Application lead (decoding, metrics, transcribe.py, Gradio Mode A, slides, demo video)
Ansasire Nehum Naamaye	QA and live-systems lead (repo, environment, CI/tests, plots, live Mode B, latency, README)

Review pairs: Data (Faith + Lisa), Model (Cole + Charis), Application (Isabelle + Nehum).

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3. Overall progress

Measure	Value
Task cards done	**21 of 26 (81%)**
Week 1 tasks	12 of 12 (100%)
Week 2 tasks	9 of 14 (64%)
Milestones closed	1 of 2 due so far (M1 done; M2 open)
Weeks 3–6 and optional	Not started

Milestone status

Milestone	Due	Status
M1 Environment & dataset verified	Oct 25	**Done** (recorded Oct 29, 4 days late)
M2 Preprocessing & alignment complete	Nov 1	**Open** (see criteria below)
M3 & M4 Baselines	Nov 8	Not started
M5 Multimodal fusion	Nov 15	Not started
M6 & M7 Core freeze	Nov 22	Not started
M9 Testing, documentation & demonstration	Nov 29	Not started

M2 completion criteria

Criterion	Status
Crop contact sheets reviewed	Marked done (W2-03), but see Risk 1
Noise-mixing tests pass	Done (W2-05)
One batch loads with consistent lengths	Done (W2-09)
Train/inference parity test passes	In progress (W2-10)
B1 full run started	Not done (blocked on W2-11 → W2-14)
Re-plan Week 3; move unfinished cards with a reason	Not done

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4. Completed work

Week 1 — Setup & Data (all done)

Card	Task	Owner / Reviewer	Due
W1-01	Kick-off meeting, roles and Trello board	Charis / Cole	Oct 19
W1-02	GitHub repository skeleton	Nehum / Isabelle	Oct 20
W1-03	Pinned Python environment	Nehum / Cole	Oct 21
W1-04	Secure GPU access and storage	Charis / Faith	Oct 21

Card	Task	Owner / Reviewer	Due
W1-05	Download and catalogue the GRID corpus	Faith / Lisa	Oct 21
W1-06	Download MUSAN and create noise pools	Lisa / Faith	Oct 22
W1-07	Dataset integrity checker (verify_dataset.py)	Faith / Nehum	Oct 23
W1-08	Speaker-independent split (splits.csv)	Faith / Charis	Oct 24
W1-09	Transcript normalisation and tokeniser	Lisa / Isabelle	Oct 23
W1-10	Mouth-cropping prototype (MediaPipe Face Mesh)	Cole / Faith	Oct 24
W1-11	Module interface specification (docs/INTERFACES.md)	Isabelle / Cole	Oct 22
W1-12	Test framework and CI (pytest + GitHub Actions)	Nehum / Charis	Oct 23
W1-M	Milestone M1	Charis / All	Oct 25

Week 2 — Preprocessing & Pipeline (9 of 14 done)

Card	Task	Owner / Reviewer	Due
W2-01	Failure handling in mouth cropping	Cole / Faith	Oct 27
W2-03	Crop quality review (contact sheets)	Isabelle / Nehum	Oct 30
W2-04	Audio standardisation (prepare_audio.py)	Lisa / Faith	Oct 27
W2-05	Noise mixer with SNR verification	Lisa / Charis	Oct 28
W2-06	Fixed noisy validation and test sets (make_noisy_test.py)	Lisa / Nehum	Oct 30
W2-07	Feature extraction module (log-Mel and MFCC)	Charis / Lisa	Oct 28
W2-09	Manifest, dataset class and collate function	Charis / Cole	Oct 30
W2-12	CTC head, greedy decoder and metrics	Isabelle / Lisa	Oct 29
W2-13	Training script (train.py)	Charis / Cole	Oct 31

Key technical specifications delivered or fixed

- **Splits:** speaker-independent. Test = 4 speakers (2F/2M), validation = 3, training = remaining (~27). `data/splits.csv` is fixed and must never be regenerated.

- **Noise:** MUSAN noise subset split by file into train/val/test pools (suggested 70/15/15, fixed seed) via `data/noise_splits.csv`, so test noise is never heard in training.
- **Mouth crops:** 96×96 grey uint8, MediaPipe Face Mesh lip landmarks, 25% margin, 5-frame moving-average smoothing, per-frame validity flag. Failed frames are interpolated, and utterances with more than 10% failed frames are flagged for exclusion. Frames are never dropped, which keeps audio–video alignment.
- **Audio features:** 25 ms Hann window, 10 ms hop, 512-point FFT, 80 Mel bands (20 Hz–8 kHz), log(x + 1e-6), trimmed to 4× video frames (300 for a 3 s clip), RMS-normalised to −25 dBFS. Config switch: `logmel80 | mfcc39 | logmel_mfcc`.
- **Noise mixer:** noise scaled by α = √(Ps / (Pw·10^(SNR/10))), measured SNR asserted within 0.01 dB of target.
- **Training script:** AdamW (lr 3e-4, wd 1e-2), 1-epoch warm-up + cosine decay, AMP, grad-clip 5.0, `CTCLoss(blank=0, zero_infinity=True)`, best-checkpoint by mean validation WER, early stopping (patience 8), resume, TensorBoard, results appended to `experiments/results.csv`.
- **CTC head:** BiLSTM (128/direction) → Linear(256→28) → log-softmax. Vocabulary is 28 symbols (blank = 0, space, a–z).

Verified results (from card comments)

Tokeniser verification (W1-09, Lisa, `scripts/verify_tokenizer.py`) on 33,000 real GRID `.align` files:

- Round-trip `decode(encode(s)) == s`: **33,000 / 33,000 pass, 0 bad**
- `encode` never emits blank
- Vocabulary: **51 unique words** observed (matches the proposal)
- Longest transcript: **31 symbols** ("place green with e eight please")
- Max adjacent repeats: **3** ("bin green by u three soon")
- CTC worst case: T ≥ L + R = 31 + 3 = 34, against T = 75 frames per 3 s clip, so the length condition is never binding
- Constants for the data loader assert: `MAX_TARGET_LEN = 31`, `MAX_REPEATS = 3`, safe minimum encoder length 34

Review sign-offs:

- W1-02: reviewed and approved on GitHub, pytest passes (Isabelle)
- W1-11: reviewed against INTERFACES.md, all checks out (Cole)

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5. Remaining Week 2 work

Card	Task	Owner / Reviewer	Due	Status	Blocked by
W2-02	Full mouth extraction (extract_mouths.py), expected ~23 GB	Faith / Cole	Oct 29	Not started	—
W2-08	Normalisation statistics (compute_stats.py)	Faith / Lisa	Oct 29	Not started	W2-02
W2-11	Audio encoder module	Cole / Charis	Oct 29	Not started	—
W2-10	Train/inference parity test	Nehum / Isabelle	Nov 1	**In progress**	W2-02
W2-14	GPU memory test and B1 overfit check	Charis / Nehum	Nov 1	Not started	W2-11
W2-M	Milestone M2	Charis / All	Nov 1	Open	all of the above

W2-14 acceptance: batch size 16 fits with AMP; B1 overfit on 8 utterances reaches CTC loss < 0.05 and CER = 0 within 300 steps; time per epoch recorded; full B1 run launched.

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6. Upcoming plan (not started)

Week	Focus	Key cards	Milestone due
3	Baselines	W3-01 evaluate B1, W3-02 evaluate.py, W3-03 visual encoder V1, W3-04 augmentation, W3-05 train B2, W3-06 train B3, W3-07 beam search, W3-08 X5 audio features, W3-09 experiment tracking, W3-10 Gradio Mode A skeleton	Nov 8 (M3 & M4)
4	Multimodal fusion	W4-01 cross-modal attention, W4-02 gated fusion and modality dropout, W4-03 fusion tests, W4-04 assemble P, W4-05 curriculum (conditional), W4-06 gate analysis, W4-07 ablation configs X1–X3, W4-08 transcribe.py, W4-09 Gradio upgrades, W4-10 live Mode B foundations, W4-11 report skeleton	Nov 15 (M5)
5	Tuning, evaluation, live demo	W5-01 LR sweep, W5-02 dropout sweep, W5-03 final runs, W5-04 ablations, W5-05 final eval and tables, W5-	Nov 22 (M6 & M7, core

Week	Focus	Key cards	Milestone due
		06 fusion analysis, W5-07 Mode B windowing/VAD, W5-08 sync calibration, W5-09 latency, W5-10 results drafts	freeze, tag v1.0-core)
6	Testing, report, demo	W6-01 to W6-10: stability tests, parity test, fresh-clone test, README, final report, slides, fallback video, rehearsal, release package, submission	Nov 29 (M9)
Optional	After core freeze	OPT-01 EfficientNet-B0 (X4), OPT-02 SNR head (X6), OPT-03 grammar-constrained decoder, OPT-04 emotion detection, OPT-05 language-ID feasibility	Nov 26–28

Final deadlines: report due Nov 27, release package Nov 28, final submission Nov 29 (card says to confirm the course deadline).

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7. Risks and issues

1. **Dependency mismatch on W2-03.** It is marked Done but depends on W2-02 (full extraction), which has not run. Contact sheets cannot yet cover the whole corpus, so the "crop review" criterion for M2 is not yet backed by full-corpus output. W2-09 and W2-10 also list W2-02 as a dependency.
2. **W2-02 is the critical bottleneck.** It blocks W2-08 and W2-10, and it was moved Done → Week 2 earlier today after being marked done in error.
3. **Three cards due today with no progress recorded:** W2-02, W2-08, W2-11.
4. **B1 is gated.** W2-14 needs W2-11 first, and Week 3's baselines depend on W2-14. No training has run, so there are no GPU memory figures or WER/CER results yet.
5. **GPU access** was the stated top project risk (W1-04). It is marked done, but `docs/compute.md` and the actual machine and GPU specs are not recorded on the board.
6. **Process gaps:**
   - W1-01 has the `docs/meeting_notes.md` task checked off, but a comment on the card says the meeting notes were not recorded.
   - M1 closed four days after its due date.
   - Many cards were moved to Done in a single bulk update, so review-before-Done (REF-01) is hard to evidence from Trello alone.
7. **Optional items** (OPT-01 to OPT-05) are lowest priority and first to drop per REF-03 if the schedule slips.

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8. Suggested items to pull from the codebase for the report

- `data/splits.csv`, `data/noise_splits.csv`, `reports/integrity_report.csv` summary: dataset statistics (usable utterances per speaker, per-check pass counts)
- Face Mesh success rate and exclusion list (from W2-02 once run), MediaPipe version and crop parameters
- Test suite results and CI status (pytest output)
- Parameter counts for encoders (the W3-03 card asks for this)
- `experiments/results.csv` and TensorBoard logs once B1 runs
- Peak GPU memory at batch size 16 and time per epoch (W2-14)
- Git history or PR list to evidence review and approval for each card

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9. Card links

Card	URL
W1-M (M1)	https://trello.com/c/45HkbQjW
W2-M (M2)	https://trello.com/c/HYqcqrS3
W2-02	https://trello.com/c/UwG0T9cM
W2-08	https://trello.com/c/etXyiHUr
W2-10	https://trello.com/c/0h0kr2xz
W2-11	https://trello.com/c/5NKePn8y
W2-14	https://trello.com/c/CFeQ6OVc
W3-M (M3 & M4)	https://trello.com/c/kBODZEMi
W4-M (M5)	https://trello.com/c/UPiKUS9a
W5-M (M6 & M7)	https://trello.com/c/9Z5PcfyZ
W6-M (M9)	https://trello.com/c/5GsqMZiT